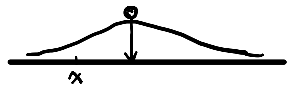
Gaussian Everywhere

① Physially Diffusion



墨水滴在细管上

$f(x,t)$ Density of Certain Particle

$$f(x,t+\tau) = \int_{-\infty}^{+\infty} \underbrace{f(x-y,t)}_{x\text{两侧的点}} \underbrace{\rho(\tau,y)}_{\text{运动过来的比率 ratio}} \, dy \quad (\text{Diffusion Integral})$$

对τ作展开：$f(x,t)+\tau\frac{\partial f}{\partial t}+o(\tau) = \int_{-\infty}^{+\infty}\left(f(x,t)-y\frac{\partial f}{\partial x}+\frac{y^2}{2}\frac{\partial^2 f}{\partial x^2}+o(y^2)\right)\rho(\tau,y)\,dy$

有：$f(x,t)+\tau\frac{\partial f}{\partial t} = \int_{-\infty}^{+\infty}\left(f(x,t)-y\frac{\partial f}{\partial x}+\frac{y^2}{2}\frac{\partial^2 f}{\partial x^2}\right)\underbrace{\rho(\tau,y)}_{\text{可视为概率密度}}\,dy \quad \cdots \quad ①$

$$\int_{-\infty}^{+\infty}\rho(\tau,y)\,dy = 1, \quad \rho(\tau,y)\geq 0, \quad \int_{-\infty}^{+\infty} y\rho(\tau,y)\,dy = 0, \quad \int_{-\infty}^{+\infty} y^2\rho(\tau,y)\,dy = D$$

因此①式可以写为：

$$f(x,t)+\tau\frac{\partial f}{\partial t} = f(x,t)+\frac{D}{2}\frac{\partial^2 f}{\partial x^2}$$

即：$\tau\frac{\partial f}{\partial t} = \frac{D}{2}\frac{\partial^2 f}{\partial x^2}$ (Diffusion Equation) $f(x,0)=\delta(x)$

解出方程：$f(x,t) = \frac{1}{\sqrt{2\pi Dt}}\exp\left(-\frac{x^2}{2Dt}\right) \longrightarrow N(0,Dt)$

② Information Theory

X 为 r.v.，熵 $H(X) = -\int_{-\infty}^{+\infty} f_X(x)\log f_X(x)\,dx$

考虑最大熵 Maximum Entropy，即最随机。在有限区间上，最先想到的为均匀分布.

在无穷区间上呢？

(1) $(-\infty,+\infty)$ $E(X)=0$ $Var(X)=\sigma^2$ 高斯分布

(2) $[0,\infty)$ $E(X)=\mu$ 指数分布

(3) $[a,b]$ $\triangleq H(f)$ 均匀分布

优化问题：$\max_f\left(-\int_{-\infty}^{+\infty} f(x)\log f(x)\,dx\right)$, s.t. $\int_{-\infty}^{+\infty} xf(x)\,dx = 0$, $\int_{-\infty}^{+\infty} x^2 f(x)\,dx = \sigma^2$

变分法 Variational Method：⟵ 将不可导变为可导的.

构造：$G(t) = H(f_0+tg)$，$t\in\mathbb{R}$，$\forall g$，其中：$f_0 = \arg\max_f H(f)$

因此 $G(t)\leq G(0)$，则 $G'(0)=0$

$$G(t) = \int_{-\infty}^{+\infty}(f+tg)\log(f+tg)\,dx$$

此时对于这个 $G(t)$，可用 Lagrange：

$$L(t,\lambda_1,\lambda_2) = \int_{-\infty}^{+\infty}(f+tg)\log(f+tg)\,dx - \lambda_1\int_{-\infty}^{+\infty}x(f+tg)\,dx - \lambda_2\int_{-\infty}^{+\infty}x^2\left((f+tg)\,dx-\sigma^2\right)$$

有

$$\frac{d}{dt}L(t,\lambda_1,\lambda_2) = \int_{-\infty}^{+\infty} g\log(f+tg)\,dx + \int_{-\infty}^{+\infty} g\,dx - \lambda_1\int_{-\infty}^{+\infty} xg\,dx - \lambda_2\int_{-\infty}^{+\infty} x^2 g\,dx$$

$$\left.\frac{d}{dt}L(t,\lambda_1,\lambda_2)\right|_{t=0} = \int_{-\infty}^{+\infty} g\left(\log f + 1 - \lambda_1 x - \lambda_2 x^2\right)dx = 0, \quad \forall g$$

则有：$\log f = \lambda_1 x + \lambda_2 x^2 - 1 \Rightarrow f = \exp(\lambda_1 x + \lambda_2 x^2 - 1) \sim N$

至此说明 (1) 为一个高斯形式.

对于 (2) (3) 是类似的.

③ Probability Central Limit Theorem (CLT)

∃一组 r.v. $X_1, \cdots, X_n$ i.i.d.假定 $E(X_k)=0$, $Var(X_k)=1$, $k=1,\cdots,n$

有 $\frac{X_1+\cdots+X_n}{\sqrt{n}} \xrightarrow{n\to\infty} N(0,1)$

Characteristic Functions : 对于 r.v. X , $\Phi_X(w) = E(\exp(jwx))$

$$= \int_{-\infty}^{+\infty} \exp(jwx) f_X(x)\,dx$$

Property : $X_1 \sim f_{X_1}(x)$, $X_2 \sim f_{X_2}(x)$ independent $X = X_1 + X_2 \sim f_{X_1} \circledast f_{X_2}$

Pf : $F_X(x) = P(X \le x) = P(X_1 + X_2 \le x)$

$$= \int_{-\infty}^{+\infty} P(X_1 + x_2 \le x \mid X_2 = x_2)\, f_{X_2}(x_2)\, dx_2$$

$$= \int_{-\infty}^{+\infty} P(X_1 \le x - x_2)\, f_{X_2}(x_2)\, dx_2$$

$$= \int_{-\infty}^{+\infty} F_{X_1}(x - x_2)\, f_{X_2}(x_2)\, dx_2$$

故 $f_X(x) = \int_{-\infty}^{+\infty} f_{X_1}(x-x_2)\, f_{X_2}(x_2)\, dx_2$

⟶ 若利用特征函数.

$$\Phi_X(w) = E\left[\exp(jw(X_1+X_2))\right]$$

$$= E\left[\exp(jwX_1)\cdot\exp(jwX_1)\right]$$

独立

$$= E\left(\exp(jwX_1)\right)\cdot E\left(\exp(jwX_1)\right)$$

$$= \Phi_{X_1}(w)\,\Phi_{X_2}(w)$$

□

考虑. $\Phi_{\frac{X_1+\cdots+X_n}{\sqrt{n}}}(w) = E\left[\exp\left(jw\frac{X_1+\cdots+X_n}{\sqrt{n}}\right)\right]$

$$= E\left[\prod_{k=1}^{n} \exp\left(jw\frac{X_k}{\sqrt{n}}\right)\right]$$

$$= \prod_{k=1}^{n} E\left[\exp\left(jw\frac{X_k}{\sqrt{n}}\right)\right] = \prod_{k=1}^{n} \Phi_{X_k}\left(\frac{w}{\sqrt{n}}\right)$$

$$\xrightarrow{\text{i.i.d}} \quad = \left(\Phi_{X_1}\left(\frac{w}{\sqrt{n}}\right)\right)^n \quad \cdots \quad ②$$

利用Taylor展开 : $\Phi_{X_1}\left(\frac{w}{\sqrt{n}}\right) = E\left[\exp\left(jw\frac{X_1}{\sqrt{n}}\right)\right]$

$$= E\left[1 + \underbrace{j\frac{w}{\sqrt{n}}X_1}_{E(X_1)=0} + \underbrace{\frac{1}{2}\left(j\frac{w}{\sqrt{n}}X_1\right)^2}_{Var(X_1)=1} + o\left(\frac{1}{n}\right)\right]$$

$$= 1 - \frac{w^2}{2n} + o\left(\frac{1}{n}\right)$$

故 ②式为 : $= \left(1 - \frac{w^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n\to\infty} \exp\left(-\frac{w^2}{2}\right)$ $\quad \left(1 + \frac{\alpha}{n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n\to\infty} \exp(\alpha)$

$$\Updownarrow$$

$\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)$ $\quad N(0,1)$

By the way , 大数定律 .

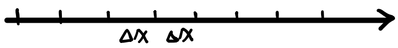
$$\frac{X_1+\cdots+X_n}{n} \xrightarrow{n\to\infty} E(X_1)$$

用特征函数看一下：

$$\phi_{\frac{X_1+\cdots+X_n}{n}}(w) = \left(\phi_{X_1}\left(\frac{w}{n}\right)\right)^n = \left(1 + j\frac{w}{n}E(X_1) + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n\to\infty} \exp\left(jwE(X_1)\right)$$

④ Stochastic Processes

考虑. Random Walk , Symmetric , 1- Dimension



$S_n = \sum_{k=1}^{n} X_n$, $X_n \sim \begin{pmatrix} \Delta x & -\Delta x \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ 两点分布

取 $X(t) = \sum_{k=1}^{n} X_k(t)$, $Var(X_k) = E(X_k^2) = \frac{1}{2}(\Delta x)^2 + \frac{1}{2}(-\Delta x)^2 = (\Delta x)^2$

$$= \left(\sum_{k=1}^{n} \frac{X_k}{\Delta x}\right) \Delta x$$

$$= \left(\frac{\sum_{k=1}^{n}\left(\frac{X_k}{\Delta x}\right)}{\sqrt{n}}\right) \cdot \sqrt{n} \cdot \Delta x$$

$$= \left(\frac{\sum_{k=1}^{n}\left(\frac{X_k}{\Delta x}\right)}{\sqrt{n}}\right) \cdot \sqrt{\frac{t}{\Delta t}} \cdot \Delta x = \left(\frac{\sum_{k=1}^{n}\left(\frac{X_k}{\Delta x}\right)}{\sqrt{n}}\right) \cdot \sqrt{t} \cdot \sqrt{\frac{(\Delta x)^2}{\Delta t}} \quad \cdots ③$$

令 $\Delta t \to 0$, $\Delta x \to 0$, $\frac{(\Delta x)^2}{\Delta t} = D$

此时 ③ $\longrightarrow N(0,1) \cdot \sqrt{Dt} = N(0, Dt)$